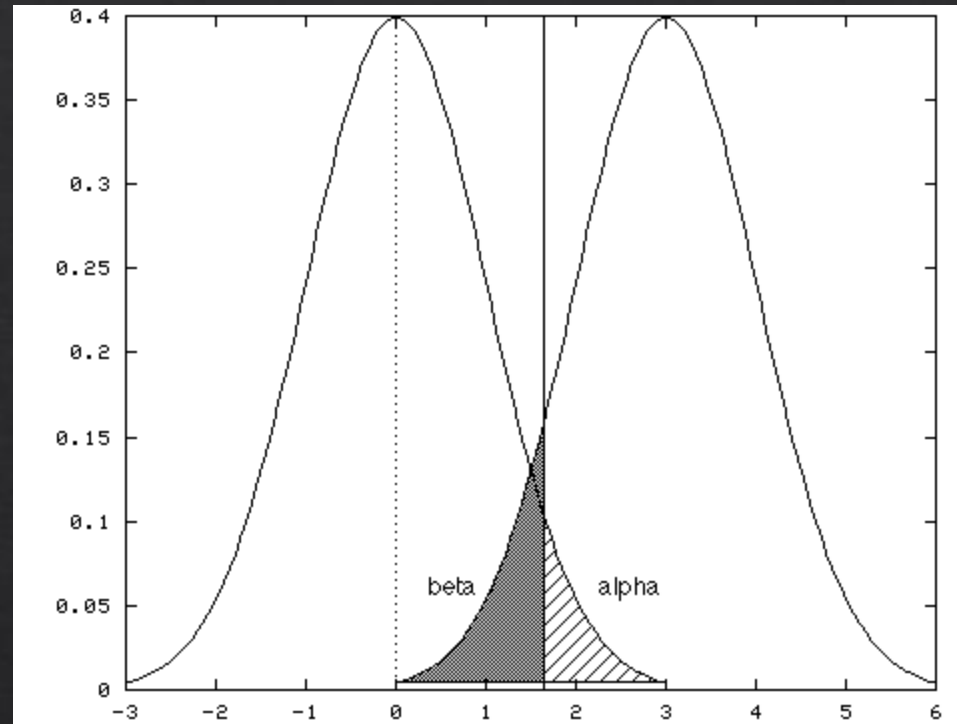




Week 8: Tests of difference; Midterm reflections

Geog4300: Harper

This week

- Hypothesis testing
- One- and two-sample tests of difference

Ingredients of classical hypothesis testing

BAKE IT SIMPLE CUPCAKES

Shopping List

- ☐ Betty Crocker Super Moist Cake Mix
- ☐ Betty Crocker Whipped Frosting
- ☐ Eggs
- ☐ Extra Light Olive Oil
- ☐ Jell-O Instant Sugar Free Vanilla Pudding
- ☐ Vanilla Extract
- ☐ Cupcake Pan
- ☐ Pretty Baking Cups
- ☐ Sparkling Sugar (pick your color)



Whip Them Up

Heat oven to 350F

IN A LARGE BOWL MIX TOGETHER:

- ☐ Cake mix
- ☐ 1 1/4 cups water (qty depends on cake flavor)
- ☐ 3 eggs
- ☐ 1/3 cup oil (qty depends on cake flavor)
- ☐ 1 teaspoon vanilla extract
- ☐ 2 tablespoons vanilla pudding powder
- ☐ Add 1/4 cup more water

MAKE IT FUN:

- ☐ Add a food color (if your making a white cake)

BAKING PAN PREP:

- ☐ Fill cupcake pan with baking cups
- ☐ Fill baking cups with cake mix (use an ice cream scoop to keep them even in size)
- ☐ Put your cupcake pan in the oven
- ☐ Set timer for 15 minutes
- ☐ When the timer goes off, check center of cupcake with a toothpick
- ☐ If toothpick comes out clean they're ready
- ☐ Cool on a rack if you have one or set on counter

Grand Finale

FROST (AFTER COOLING 1 HOUR):

- ☐ Frost with the back of a spoon or frosting tip if your adventurous
- ☐ Add sparkling sugar to the top (this is for the Glitter Girl in all of us)
- ☐ Put your cupcakes on a pretty platter or cake stand to display your wonderful creations

SHARE THIS DELIGHTFUL TREAT WITH YOUR FAMILY, FRIENDS, OR CO-WORKERS!

glitterGUIDE

Steps in hypothesis testing

1. State the null and alternate hypothesis

Null hypothesis (H_o):

A finding of **no difference**

$$\mu = \mu_H$$

- Focuses on the assumption that no differences exist.
- Involves an = sign.

Alternative hypothesis (H_A):

A finding of **inequality**

$$\mu \neq \mu_H$$

Two-tailed test

$$\mu > \mu_H$$

One-tailed test

$$\mu < \mu_H$$

Steps in hypothesis testing

1. State the null and alternate hypothesis
2. Select the appropriate test

Tests of difference

What is the specific purpose of this test? When would you use it rather than another one?

- One sample t-test
- Two sample t-test
- Two sample paired t-test
- Non-parametric (Wilcoxon) t-test
- Difference test between two proportions

Tests of difference

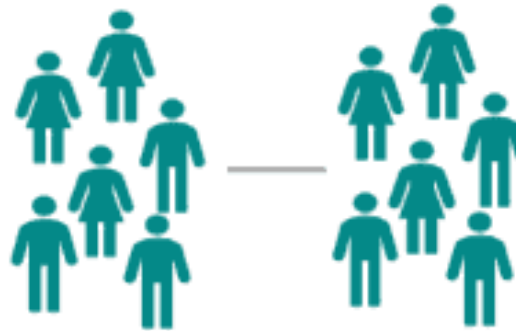
Is there a difference?

One sample t-Test



Is there a **difference** between a **group** and the **population**

Independent samples t-Test



Is there a **difference** between **two groups**

Paired samples t-Test



Is there a **difference** in a **group** between **two points in time**

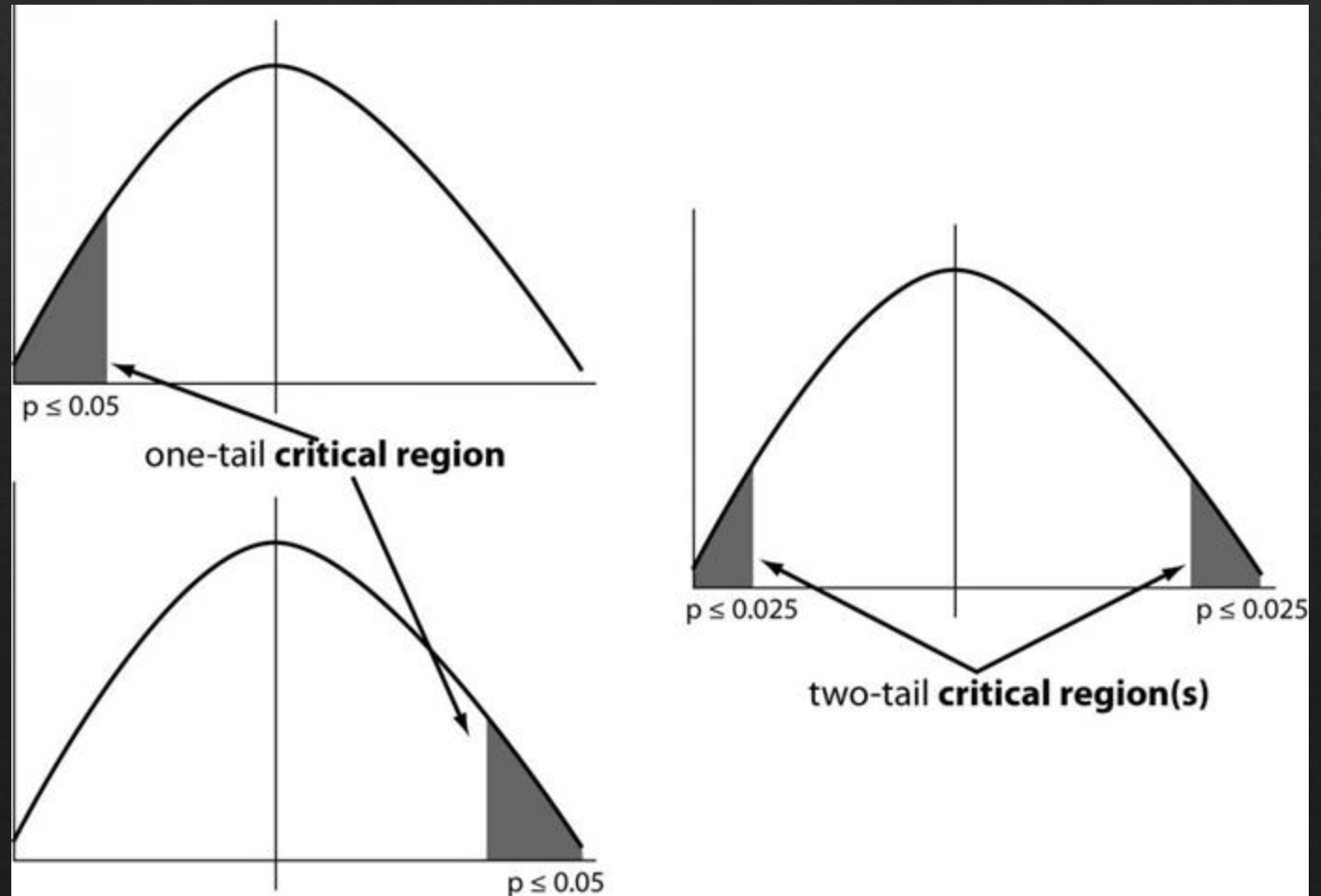
Steps in hypothesis testing

1. State the null and alternate hypothesis
2. Select the appropriate test
3. Select the level of significance

Select level of significance

$P < 0.05$:

The probability that this sample comes from a population in which the two means are equal is less than 5%



Steps in hypothesis testing

1. State the null and alternate hypothesis
2. Select the appropriate test
3. Select the level of significance
4. Delineate regions of rejection and non-rejection of the null hypothesis

Delineate regions of rejection / non-rejection

Degrees of freedom:

One sample test: $n-1$ (24 in our case)

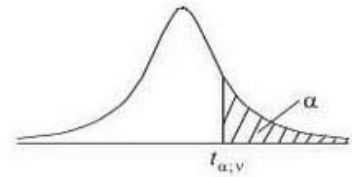
Critical value:

Based on confidence level (alpha) and df

Table of the Student's t -distribution

The table gives the values of $t_{\alpha;v}$ where

$\Pr(T_v > t_{\alpha;v}) = \alpha$, with v degrees of freedom



$\alpha \backslash v$	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Delineate regions of rejection / non-rejection

Degrees of freedom:

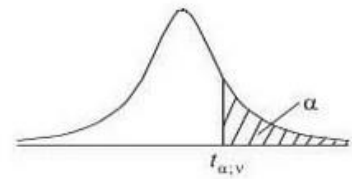
One sample test: $n-1$ (24 in our case)

Critical value:

Based on confidence level (alpha) and df

Table of the Student's t -distribution

The table gives the values of $t_{\alpha;v}$ where
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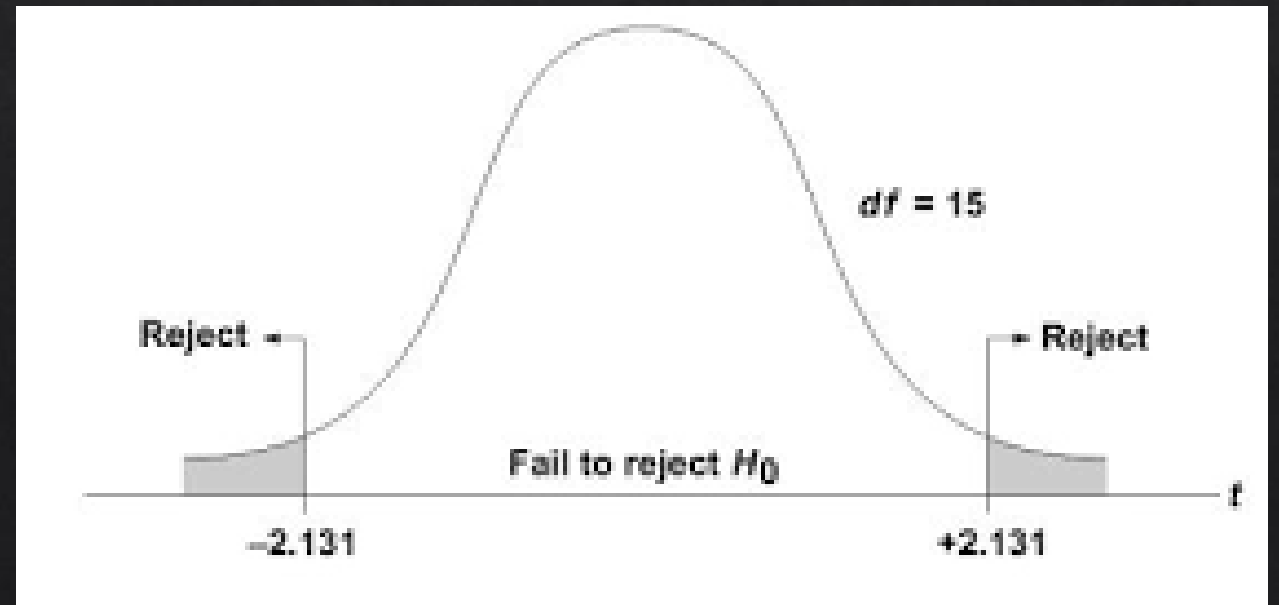


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4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
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∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Reject or fail to reject?

Based on test results, we *reject* or *fail to reject* the null hypothesis.

We don't prove or disprove it, but show it is likely or unlikely to be true.



Steps in hypothesis testing

1. State the null and alternate hypothesis
2. Select the appropriate test
3. Select the level of significance
4. Delineate regions of rejection and non-rejection of the null hypothesis
5. Calculate the t-statistic

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Steps in hypothesis testing

1. State the null and alternate hypothesis
2. Select the appropriate test
3. Select the level of significance
4. Delineate regions of rejection and non-rejection of the null hypothesis
5. Calculate the t-statistic
6. Make a decision

Hypothesizing

Give the null and alterative hypothesis for the following research questions.

- Is the average commute time higher on Mondays than it is the rest of the week?
- Is the amount of property damage from severe weather different between southern states?
- Is there an association between college teams' football records and rates of student enrollment?
- Is the rate of flu cases low in our area compared to our state?

Let's try it out

The American Community Survey indicates that statewide, the average commute time is 22 minutes

We surveyed 36 community members:

- Mean commute: 17.2 minutes
- Standard deviation: 3.2 minutes

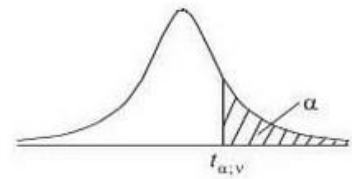
Is our group different from the state average?

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

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Midterm reflection

Revisiting the grading rubric:

	A	B	C	D
<i>Labs submitted</i>	All submitted	All submitted	At least five submitted	At least three submitted
<i>Labs marked complete</i>	All complete	At least five complete	At least three complete	At least two complete
<i>Labs needing revision</i>	No more than three	No more than four	No more than five	NA
<i>Labs submitted late</i>	No more than two	No more than two	No more than four	No more than four
<i>Attendance</i>	>75% of classes	>65% of classes	>50% of classes	>40% of classes
<i>Challenge items</i>	Complete three or more	Complete two or more	Not required	Not required

Student 1:

Submitted Lab 0, Lab 1, and Lab 2, but Lab 1 was incomplete. It has been revised. Has attended 80% of classes and completed one challenge question.

Midterm reflection

Revisiting the grading rubric:

	A	B	C	D
<i>Labs submitted</i>	All submitted	All submitted	At least five submitted	At least three submitted
<i>Labs marked complete</i>	All complete	At least five complete	At least three complete	At least two complete
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<i>Labs submitted late</i>	No more than two	No more than two	No more than four	No more than four
<i>Attendance</i>	>75% of classes	>65% of classes	>50% of classes	>40% of classes
<i>Challenge items</i>	Complete three or more	Complete two or more	Not required	Not required

Student 2:

Submitted Lab 0, Lab 1, and Lab 2, but Lab 1 and 2 were both incomplete. Lab 1 has been revised. Has attended 90% of classes and completed two challenge questions.

Midterm reflection

Revisiting the grading rubric:

	A	B	C	D
<i>Labs submitted</i>	All submitted	All submitted	At least five submitted	At least three submitted
<i>Labs marked complete</i>	All complete	At least five complete	At least three complete	At least two complete
<i>Labs needing revision</i>	No more than three	No more than four	No more than five	NA
<i>Labs submitted late</i>	No more than two	No more than two	No more than four	No more than four
<i>Attendance</i>	>75% of classes	>65% of classes	>50% of classes	>40% of classes
<i>Challenge items</i>	Complete three or more	Complete two or more	Not required	Not required

Where are you currently?

Midterm reflection

1. What are two things you have learned about data science and quantitative analysis that are NOT about learning R/coding?
2. What is at least one thing you have found valuable about your work in R/Rstudio/Github?
3. What resources have been most useful for you in completing assignments in this course? How have these been helpful? These could include classmates and/or online options such as ChatGPT or StackOverflow.
4. What do you want to gain or improve on as a learner from the remainder of this course?
5. Are there any changes I could make to the course to help you as a learner?

Submit via the ELC Assignments tool

How are proportions different?

The *standard error* is different for proportions/rates.

We also use a z statistic rather than a t-statistic

One sample T-test for counts

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

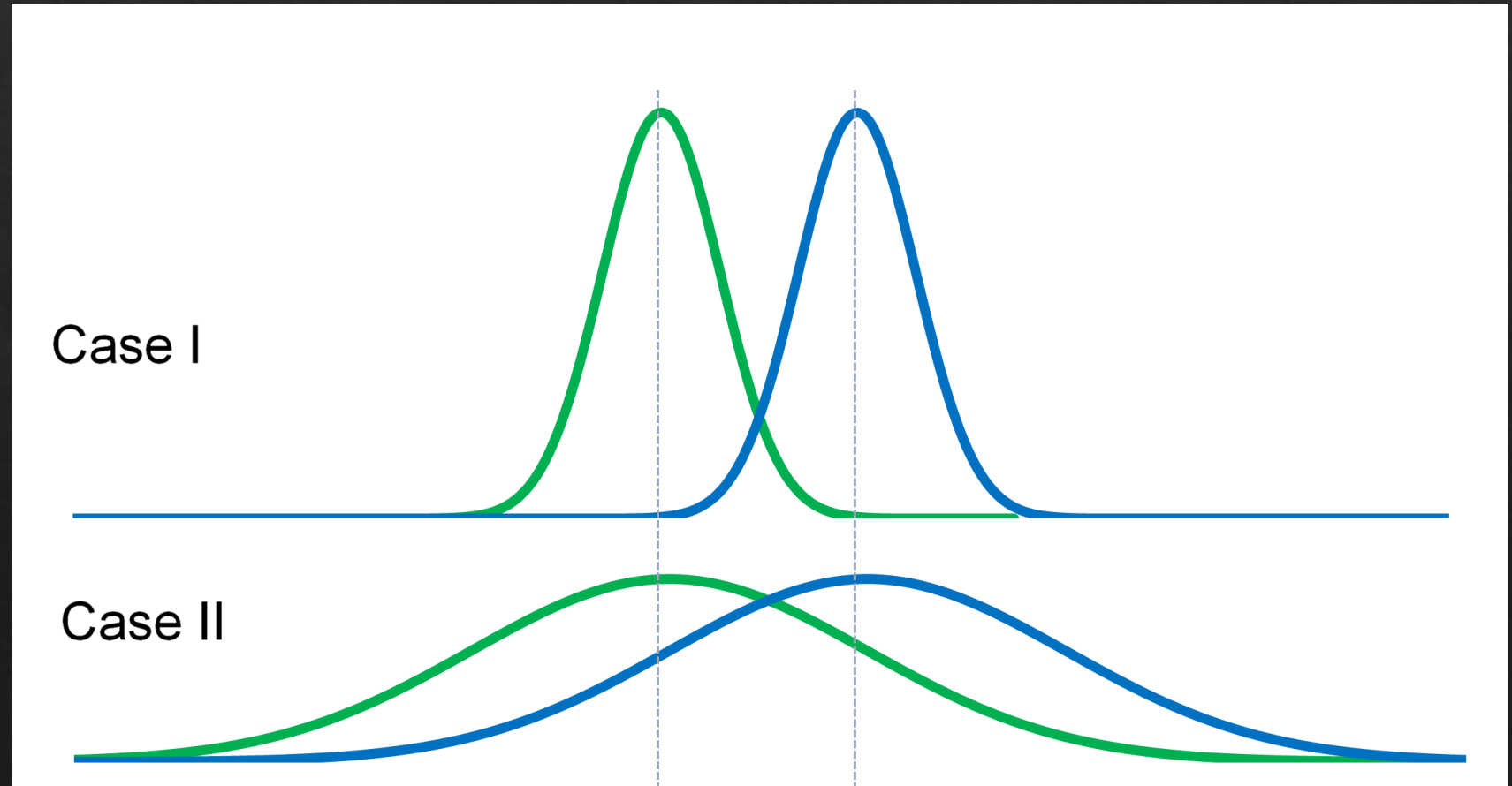
Z-test for proportions

$$z = \frac{\hat{p} - \rho}{\sqrt{(p * (1 - p)) / n}}$$

Two sample t-tests

Instead of comparing one sample against a predetermined mean, identifying difference between two samples

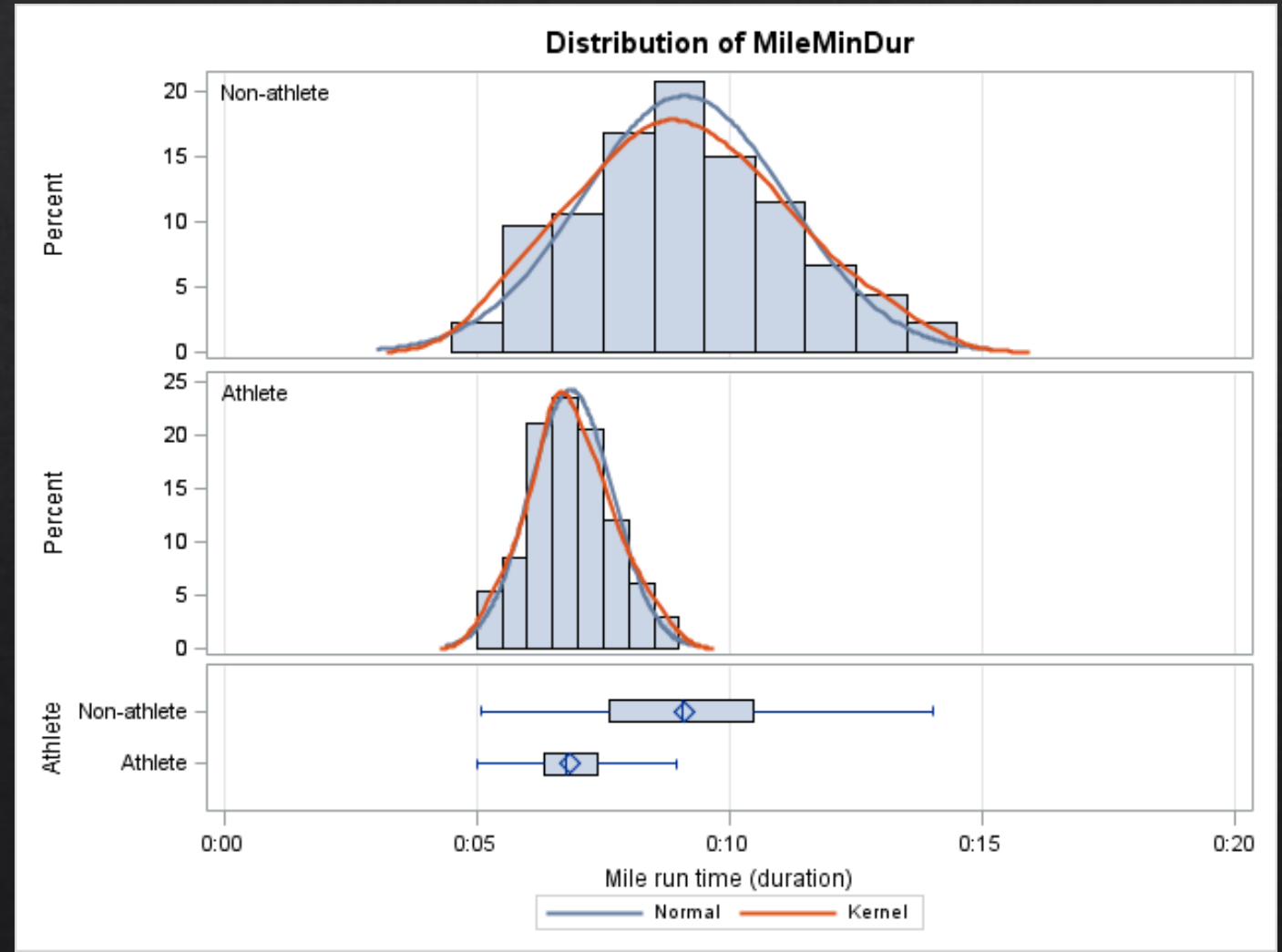
$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$



Two sample t-tests

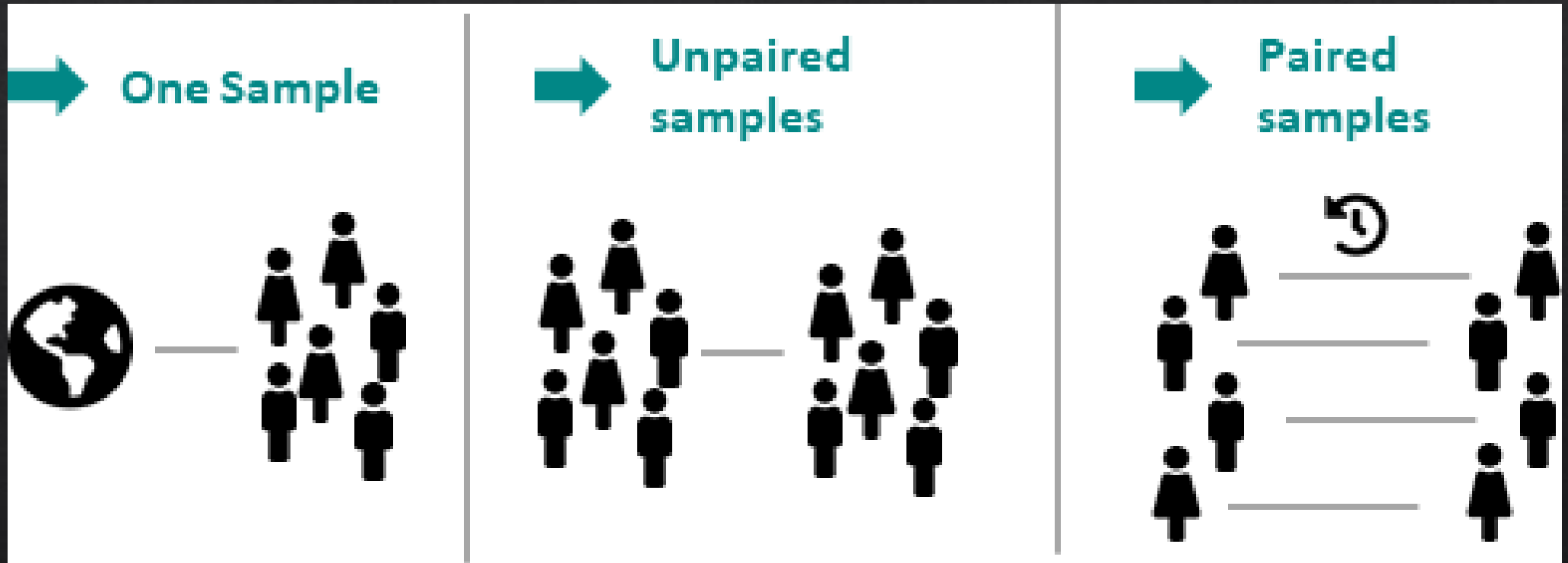
Key question #1: Is the variance in both samples *even* or *uneven*?

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$



Two sample t-tests

Key question #2: Are the samples *independent* or *paired*?



Are the samples *independent* or *paired*?

On a sheet of paper, write down two research questions for your area of interest. One should use two independent samples and one should use a paired sample.

